STRATEGY OVERVIEW · NQ / MNQ

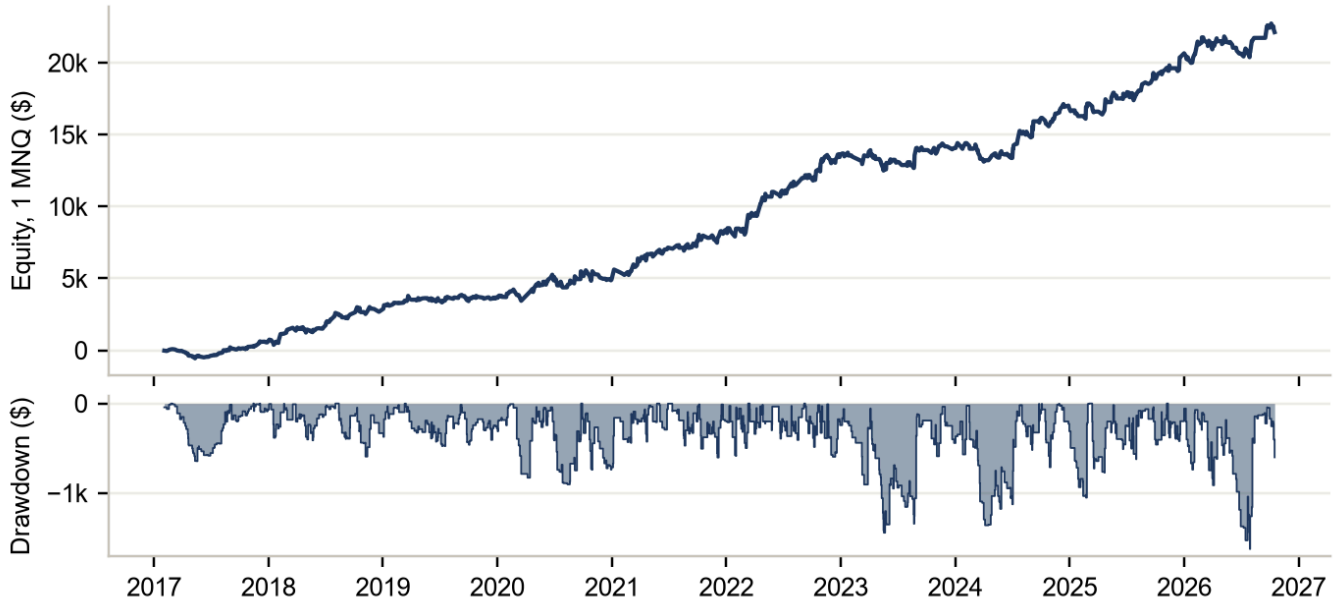
FC 2B — Flow-Confirmed Opening Drive

E-mini Nasdaq-100 · 1-minute bars · day session 08:30–15:00 CT · one trade a day, long or short · flat by 14:59 CT

Net, 1 MNQ	Profit factor	Trades a year	Win rate	Max drawdown	Robustness grade
$22,126	1.51	88	59.9%	$1,627	B

MultiCharts backtest, 2017-02-02 → 2026-10-16, 856 trades, costs included, 1 MNQ (×10 = 1 NQ). Max drawdown counts closed equity plus each trade's worst open loss (MAE). Closed-trade drawdown $1,444.

Equity path and drawdown



Worst drawdowns (including open-trade loss)

Depth, 1 MNQ	Peak	Low	Recovered	Days
$1,627.25	2026-05-08	2026-07-29	2026-09-21	136
$1,444.75	2023-04-05	2023-05-18	2023-08-29	146
$1,361.75	2024-01-08	2024-04-05	2024-07-18	191
$1,052.75	2024-12-11	2025-02-21	2025-02-25	75

Depth ×10 = 1 NQ. Days = calendar days from peak to recovery. Now −$602.00 from the peak.

By year

Year	Trades	1 MNQ	1 NQ
2017	83	$616.00	$6,160
2018	83	$2,060.50	$20,605
2019	97	$958.00	$9,580
2020	83	$1,281.50	$12,815
2021	85	$3,436.00	$34,360
2022	95	$5,244.50	$52,445
2023	82	$468.00	$4,680
2024	91	$2,752.50	$27,525
2025	89	$3,841.00	$38,410
2026	68	$1,468.00	$14,680

2026 to 2026-10-16.

The trade story

Every morning it watches the first 30 minutes of the New York session (08:30–09:00 CT). If that window closes up or down **and** the order flow inside it was unusually one-sided the same way compared with its own last 60 days, it enters that direction at 09:01 and holds into the afternoon with a wide stop, a break-even step and a loose trail — no target. Most days give back a little; the trend days pay for everything.

Rules

Opening range	1-minute bars stamped 08:31–09:00 CT.
Direction	Sign of the range's close minus its open.
Flow filter	Per bar Volume × (Close − Open) ÷ (High − Low), summed over the range ÷ range volume; scored as a z-score against the previous 60 days (≥ 20 needed). Trade only if direction and z agree beyond ±1.0.
Entry	Market at the 09:01 bar.
Exits	Stop 100 points · after +66 points the stop moves to entry + 1 point · 150-point trail from the best price · no target · market exit from 14:58 CT.

Who drives it, and the evidence

The idea

An opening drive backed by one-sided order flow is more likely to become a trend day than one that is not. The flow z-score is the filter; the wide stop, loose trail and missing target let the trend days run.

Inspired by Zarattini & Aziz, "Can Day Trading Really Be Profitable?" (SSRN 4416622, opening-range breakout on QQQ), but structurally different: 30-minute instead of 5-minute range, no breakout level, an order-flow filter, no 10R target, and a ~60% win rate instead of the paper's 24%.

Is the edge in the signal or in the hours?

Random-entry benchmark on the 2023-02 → 2026-09 trades (302): the strategy made **$7,101.50**; the same clock hours on random days made **−$561.50** (p 0.046). A random direction on the same days loses (p 0.024). Shorts made $3,284.50. On the 2017+ trades (in sample) p 0.0008, and it beats the benchmark in all 10 years, including 2022.

Remove the flow gate entirely and, 2017–2022, PF drops from 1.60 to 1.21 while max drawdown triples ($1,040.50 → $3,189).

Robustness grade B

Split at 2023-01-01, 1 MNQ	2017–2022	2023 → (later period)
Trades	526	330
Expected value per trade	$25.85	$25.85
Kept in 2023 →		100% of the 2017–2022 edge · P(edge ≤ 0) = 0.0126
2023 → net / max drawdown		$8,530 / $1,444 (closed) · return ÷ drawdown 5.91
Monte-Carlo max drawdown		median $2,119 · 75th $2,572 · 95th $3,508 · 99th $4,393

How it was validated: the first 2B rule — stop at the opening-range extreme, break-even step, no trail — was fitted on 2016–2021 (455 trades, PF 1.51, net $7,570.50), checked on 2022–2023 (PF 1.72, net $7,560) and run once, frozen, on the Feb 2024 → 2026-06-01 holdout: 211 trades, PF 1.42, net $7,106.50, max drawdown $2,320.50 (1 MNQ). That holdout was later re-read to reject variants and then retired. Today's exits — a fixed 100-point stop and a 150-point trail — were set on 2026-07-19 on the full sample. They give up some profit for a smaller drawdown (about $20,640.50 / $1,611 against the first rule's ~$21,900 / $2,320.50 per MNQ) and were registered as a forward challenger. The 2023 → column is therefore a later-period check of today's rules, not a frozen out-of-sample run.

Expected value from a 10,000-draw bootstrap; drawdowns from a 20,000-path Monte Carlo of the trade sequence (closed equity). Risk unit = 2017–2022 average loss ($101.04 per MNQ). Grade scale S–D combines the share of the 2017–2022 edge kept in 2023 →, the confidence that the 2023 → edge is above zero, and return ÷ drawdown.

Recent window

Backtest 2023-02-03 → 2026-10-09: 87 trades a year · net $9,561.5 · PF 1.48 · average $29.79 · max drawdown $1,627.25 (closed $1,444.00) · last 3 months $2,087.5.

Earlier MultiCharts runs, 1 MNQ

Window	Trades	Net	PF	Win rate	Average	Max strategy DD
2017-02-02 → 2026-07-01 (full sample)	830 (403 L / 427 S)	$20,640.50	1.49	59.64%	$24.87	$1,611
2017-02-02 → 2022-12-30	526	$13,596.50	1.60	57.4%	$25.85	$1,040.50
2023-02-03 → 2026-09-01	312 (148 L / 164 S)	$8,762.50	1.45	64.1%	$28.10	$1,864

Separate runs: the z-score history restarts at each run's start. All three use today's exits (set 2026-07-19), so 2023 → is a later-period check, not a frozen out-of-sample run; the one holdout run is described under the robustness grade. Full sample: average win $125.98, average loss −$124.91, largest win $885.50, largest loss −$385, time in market 4.25%, average hold ≈ 4 h 17 min, t-statistic 4.02.

> **The strategy is its tail**
>
> Full sample 2017-02 → 2026-07: **75 trades (9%) made $30,390.50** (about $405.20 each); the other 755 lost −$9,750 (about −$12.90 each). The same shape holds in both halves. That is why there is no target and the trail is loose: a target caps exactly the days that pay for everything.

Tried and rejected — do not repeat

When	Tried	Result (1 MNQ)	Verdict
2026-08-11	Profit target 200 points (with three entry filters)	Net $14,160 vs $20,640.50; largest win $399 vs $885.50 — the target caps the tail	Never add a target
2026-08-11	Location of price inside the opening range	Worse	Rejected
2026-08-11	Late-range commitment (last 10 minutes' volume and delta)	Worse	Rejected
2026-08-11	Break-then-retest entry with pullback quality	Worse	Rejected
2026-10-06	ATR regime filter (ATR20 ÷ ATR100), 2017–2022	Every cut loses net at the same drawdown	Rejected
2026-10-06	Opening-range width ÷ ATR20 cap	Best cap ≤ 0.6: PF 1.68 but net $11,736 vs $13,596.50	Rejected
2026-10-06	ATR position sizing	Inverse ATR: net $8,801.70, DD $1,752.50. Proportional: net $18,951.20 but more drawdown and lower t — leverage, not edge	Rejected
2026-10-06	ATR-scaled stop (a × ATR20) with matching trail	No multiple beats 100 / 150 points	Rejected
2026-10-06	Time stops at 10:00 / 10:30 / 11:00 / 12:00	7 of 8 worse; the one better is noise	Rejected
2026-10-06	Add a contract at +66 / +100 / +150 points	+66: net ÷ drawdown 16.4 vs 15.3 — mostly extra size	Not adopted
2026-10-06	Exit at 15:08 or 15:13 instead of 14:58	Both worse	Rejected
2026-10-06	Overnight (Globex) position agreeing with the trade, as a filter	Both halves profitable, so filtering loses money	Rejected
2026-10-11	London agreement, London direction as the signal, London ADX gate	Agree vs disagree not significant (p 0.43); London direction alone PF 1.04; ADX-gated days did worse (PF 1.50 vs 1.70)	Rejected
2026-07-11	The source paper's own structure: 5-minute range, stop at the range extreme, 10R target	Available as a setting in the first version; no result recorded	Open

Why filters fail here: nothing in the opening range predicts which day becomes a big winner, so every filter removes winners with the losers. 2017–2022 screens are Python replays that reproduced the MultiCharts exits trade for trade; 2023+ was not used for them.

Build history — every version built, in order

Built	What changed	Result (1 MNQ)	Verdict
2026-07-11	First version: 5-minute opening range, stop at the range extreme, optional R-multiple target, flow filter on NQ alone or NQ + ES, and a no-filter breakout control	Not recorded	Replaced
2026-07-12	Break-even step: after +0.66 R the stop moves to entry + 4 ticks (fit window to 2021)	Sensitive: 0.5 fine, 0.75 collapses; 0.66 kept as the registered value	Adopted
2026-07-12	Removed after testing: ES cross-flow (redundant with NQ's own flow), VWAP at entry, VWAP-cross exit, flow-flip exit, ATR-fraction stop, tick up/down delta, minimum-stop filter	ES cross-flow changed 8 of 208 trades; the VWAP-cross exit failed the holdout; other figures not recorded	Rejected
2026-07-12	Flow z ≥ 2	Under 100 trades — too few to rank	Rejected
2026-07-12	Flow z ≥ 1.75 with a 70-day lookback; one break-even variant	Best on the development data, worse on the holdout	Rejected
2026-07-12	First 2B rule: 30-minute range, flow z ≥ 1.0 against the last 60 days, stop at the range extreme, break-even 0.66 R, no target	2016–2023: $15,130.50. Holdout Feb 2024 → 2026-06-01: 211 trades, PF 1.42, net $7,106.50, max drawdown $2,320.50	Adopted
2026-07-12	A break-even that leaves part of the risk on	Option, off by default; not recorded	Not adopted
2026-07-13	Profit target at 2.7 R	2016–2023: $2,150 better than 2B. Holdout: $830 worse, PF 1.37 vs 1.42, same max drawdown	Rejected
2026-07-13	$200 dollar stop across 2 NQ	A 5-point stop on ~77% of trades; largest loss $7,670 (2 NQ) went through the cap, since the entry bar has no working stop	Rejected
2026-07-13	Afternoon trade: same rule on a 12:30–13:00 range, held to the close	2016–2023: 598 trades, $773, PF ≈ 1.05. 2024–2026: 194 trades, −$749.50, PF ≈ 0.92	Rejected
2026-07-13	Morning trade exited at 12:28 instead of the close	2016–2023: $10,874 vs $15,130.50 held to the close. Holdout: $7,919 vs $7,106.50, PF 1.51, deeper drawdown ($2,489.50 vs $2,320.50)	Not adopted; forward log only
2026-07-13	Same rule at the Asia (17:00) and Europe (02:00) opens, one trade each, then two legs per session (2016–2023)	The four overnight legs together ≈ −$9,500 over ~3,350 trades (−$2.80 a trade)	Rejected
2026-07-14	Hourly legs (12 a day across sessions); pyramids adding up to three agreeing legs — shared or own stops, 1:1 targets, 1-2-3-4-contract ladders, US only, stop at half the distance to the range extreme	Not recorded	Not adopted
2026-07-14	5-minute bars instead of 1-minute (6-bar z-score), with a $900 cap	Per-trade Sharpe 0.06 vs 2B's 0.117; largest loss $3,950 went through the cap	Rejected
Jul 15–19	Fixed targets of 90, 150 and 300 points on a 100-point stop, with and without break-even	Losses unchanged; profit rises with the target. A 350-point target without break-even made $21,875.50 ≈ 2B's net	Rejected — converges back to 2B
Jul 15–19	50-point stop with break-even at +33 points	Net $15,000–$17,100, max drawdown $1,270–$1,360: 70–78% of 2B's net	Forward-trial option only

Jul 15–19	Time exit only: out a set number of minutes after entry; no stop, target or trail	Not recorded	Not adopted
2026-07-19	Fixed 100-point stop, break-even at +66 points and a 150-point trail in place of the range-extreme stop, set on the full 2017–2026 sample	Today's FC 2B (results above). Best return ÷ drawdown of the exit tests (12.8 vs 9.6) but less net than 2B, so it missed its pre-set "beats 2B" bar	Forward challenger; adopted before its 30-signal forward test finished
2026-08-11	Fixed stop and fixed target without trail or break-even, alone and with the three entry filters	Not recorded	Not adopted
2026-08-11	Session-VWAP trailing stop	Failed; figures not recorded	Rejected
2026-08-11	Take profit at the 2σ VWAP band, re-entries at the 1σ band, no trail or break-even	Failed; figures not recorded	Rejected
2026-08-17	Gold (COMEX) on London hours: 05:30–06:00 ET range, 20-point stop; with the trail, and with a fixed target instead	Unprofitable; figures not recorded	Not adopted

"Not recorded" means no figures were kept for that build; it is listed so the same build is not repeated blind. The 2026-08-11 target and filter tests and the 2026-10 screens are in the table above. Exit tests, July 2026: every exit tried lies on one line trading net profit against drawdown (2B ≈ $21,900 / $2,320.50 → trail $20,640.50 / $1,611 → 50-point stop ≈ $15,000–$17,100 / $1,270–$1,360); none beats the first 2B rule on both.

After 2024 — does the regime still show up?

So far yes: $2,752.50 in 2024, $3,841.00 in 2025 and $1,468.00 in 2026 to 2026-10-16 (1 MNQ). In the 2024–2026 part of the full run, 239 trades averaged $32.10 with a 66.5% win rate and PF 1.50. One thing has changed: the fixed 100-point stop was 2.2 × daily ATR in 2017 and is about 0.23 × ATR in 2026 — tighter, in volatility terms, than anything in the development years.

Kill test — what would prove it wrong

Watch	From the data	Warning sign
Drawdown	Worst so far $1,627 (incl. open loss); Monte-Carlo 75th percentile $2,572 (closed equity)	Beyond the 75th percentile.
Win rate	59.9%; it needs 49.8% to make money (average win $127.53, average loss −$126.61)	Win rate sliding toward 49.8%.
Trade count	≈ 88 a year	Materially more or fewer: setup drift.
Control	Signal beats a random direction and the same hours on random days	Profit no better than random days at the same hours.

Long 416 trades $11,820 · short 440 trades $10,306 · largest win $885.50, loss −$385.00 · median hold 358 min · positive months 75 of 117. Source: MultiCharts backtests; rules from the strategy code. Backtests, not account results. Prepared 2026-10-22.